

Prompt Engineering Mini Project

Designing a Smart College Library Assistant

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1. Problem I Chose

Students often ask the library for help with books, due dates, availability, and renewals. A simple AI assistant can understand a student's message, identify what the student needs, and suggest the right next action. In this mini project, I use prompt engineering to make that process more organized.

2. Sample Student Message

"Hi, I borrowed a Data Structures book last week. I have an internal exam on Monday, so I want to keep the book for a few more days. Can you tell me whether I can renew it?"

3. My Prompt Design

Instead of asking the AI to do everything in one step, I give it three focused jobs. First it identifies the student's request. Then it checks the situation against simple library rules. Finally it prepares a helpful response for the student.

Stage A — Understand the request

Prompt:

Read the student's message carefully. Identify the book mentioned, the reason for the request, the requested action, and any important date. Do not add information that the student did not say.

Expected result:

Book: Data Structures

Reason: Internal exam on Monday

Requested action: Renew the borrowed book

Important date: Monday

Why this helps

This first prompt removes unnecessary words from the student's message and keeps only the details needed for the next decision. The result is easy for another prompt to use.

Making the Decision

4. Stage B — Apply Simple Library Rules

Now the AI takes the details from the first stage and compares them with a small set of rules. These rules are intentionally simple so that the decision is easy to explain.

Library Rules

1. A student can request a renewal when another student has not reserved the book.
2. A renewal can be suggested for a short additional period, such as seven days.
3. If the book has already been reserved by someone else, the student should return it on time.
4. When the rules do not give enough information, the assistant should ask the student to check with the library.

Decision Prompt

Use the information from Stage A. Apply the library rules one by one. Decide whether renewal can be recommended from the information available. Give a clear reason. Do not assume that the book is reserved unless the message says so.

Decision

Recommendation: Renewal can be requested.

Reason: The student wants extra time because of an upcoming internal exam. Nothing in the message says that another student has reserved the book, so there is no stated reason to reject the request. The library can check the reservation status and then approve or decline the renewal.

5. Stage C — Prepare a Student-Friendly Reply

The last stage turns the decision into a natural reply. The answer should sound like a helpful library assistant, not like a technical system message.

Response Prompt

Write a short and polite reply to the student. Mention the book, the reason for the renewal request, and what the student should do next. Keep the language simple and do not claim that the renewal has already been approved.

Draft Reply

Hi! You can request a renewal of the Data Structures book because you need it for your internal exam. The library will first need to check whether another student has reserved the book. If it is not reserved, your renewal request can be approved. Please check the library portal or contact the library desk to confirm.

Final Output and Reflection

6. Structured Result

For record keeping, the useful information can also be stored in a structured format. This makes the output easier for a website or library application to process.

```
{
  "student_request": "Renew borrowed Data Structures book",
  "reason": "Internal exam on Monday",
  "book": "Data Structures",
  "current_decision": "Renewal can be requested",
  "condition": "Library must check whether the book is reserved",
  "next_action": "Check the library portal or contact the library desk"
}
```

7. Complete Workflow at a Glance

Part	What the AI does	Result
Understand	Finds the book, reason, requested action, and date.	Key details are separated from the original message.
Decide	Checks the details against the library rules.	Renewal can be requested, subject to reservation status.
Respond	Writes a polite and practical reply.	Student receives clear next steps.
Structure	Stores the result in a fixed data format.	Information can be used by a library system.

8. What I Learned

This project helped me understand that a good prompt does not have to be complicated. Breaking a real problem into smaller tasks makes the AI response easier to control. The first task focuses on understanding, the second on decision-making, and the third on communication. Using a structured output at the end also makes the result useful for a software application.

9. Final Note

The main idea of this mini project is to make an AI assistant behave in a clear and predictable way. The prompts are written in everyday language, while the final structured result can be understood by both people and computer programs.